

Homework-4

1. **Photoionization of a hydrogen atom:** Consider a hydrogen atom is irradiated by a high frequency (ω) electromagnetic radiation which is described by the vector potential

$$\mathbf{A}(\mathbf{r}, t) = A_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \hat{\mathbf{e}}.$$

Here A_0 is the amplitude of the vector potential and $\hat{\mathbf{e}}$ is the polarization vector of the field. In this photoionisation process, an electron is ejected from a hydrogen atom. Assume this atom is initially in the ground state described by $\psi_{100}(r)$ with energy $E_1 = -13.6$ eV. Energy of the absorbed photon is $E = \hbar\omega$, where $\omega = kc$ with k is the wave vector of the photon. Let $\mathbf{k}_f$ be the wave vector of the electron in the final continuum states and $\mathbf{p}_f = \hbar\mathbf{k}_f$ is the momentum. Kinetic energy of the final state is $E_f = \hbar^2 k_f^2 / 2m \ll mc^2$.

- (a) Show that the absorption rate for the transition from the initial state to the final continuum states is

$$\Gamma = \frac{2\pi}{\hbar} \left(\frac{eA_0}{2m} \right)^2 \sum_{\mathbf{k}_f} |\langle f | e^{i\mathbf{k} \cdot \mathbf{r}} \hat{\mathbf{e}} \cdot \mathbf{p} | i \rangle|^2 \delta(E_f - E_i - \hbar\omega),$$

where $\mathbf{p} = -i\hbar\nabla$.

- (b) Show that the matrix element M_{fi} is

$$M_{fi} = \langle f | e^{i\mathbf{k} \cdot \mathbf{r}} \hat{\mathbf{e}} \cdot \mathbf{p} | i \rangle = \frac{\hbar k_f}{L^{3/2}} \frac{8\pi a_0^3}{\sqrt{\pi a_0^3}} \frac{\hat{\mathbf{e}} \cdot \hat{\mathbf{k}}_f}{(q^2 a_0^2 + 1)^2},$$

where $q = |\mathbf{k} - \mathbf{k}_f|$.

- (c) Show that in the non-relativistic limit,

$$\begin{aligned} \frac{k}{k_f} &= \frac{v_f}{2c} \ll 1 \\ q^2 &= k_f^2 \left(1 - \frac{v_f}{c} \cos \theta + \left(\frac{v_f}{2c} \right)^2 \right) \sim k_f^2 \left(1 - \frac{v_f}{c} \cos \theta \right) \sim k_f^2 \\ k_f a_0 &\gg 1. \end{aligned}$$

Here we have assumed that wave is polarized along x axis and θ is the angle between $\hat{\mathbf{k}}$ and $\hat{\mathbf{k}}_f$.

- (d) Using

$$\sum_{\mathbf{k}_f} \rightarrow \frac{L^3}{(2\pi)^3} \int d^3k,$$

show that the absorption rate per unit solid angle $d\Omega$ is

$$\frac{d\Gamma}{d\Omega} = \frac{16e^2 A_0^2 (k_f a_0)^3}{4\pi m \hbar} \frac{(\hat{\mathbf{e}} \cdot \hat{\mathbf{k}}_f)^2}{(k_f^2 a_0^2 + 1)^4} = \frac{16e^2 A_0^2 (k_f a_0)^3 \sin^2 \theta \cos^2 \phi}{4\pi m \hbar (k_f^2 a_0^2 + 1)^4},$$

where $k_f^2 = \frac{2m\omega}{\hbar} - \frac{2m|E_1|}{\hbar^2} \simeq \frac{2m\omega}{\hbar}$.

(e) Show that the total absorption rate is

$$\Gamma = \frac{16e^2 A_0^2 (k_f a_0)^3}{4\pi m \hbar} \frac{1}{(k_f^2 a_0^2 + 1)^4} \int \sin^2 \theta \cos^2 \phi d\Omega = \frac{16e^2 A_0^2 (k_f a_0)^3}{3m \hbar} \frac{1}{(k_f^2 a_0^2 + 1)^4},$$

(f) Absorption cross section is defined as

$$\sigma_{abs} = \frac{\text{energy} \times \text{absorption rate}}{\text{energy flux}}.$$

Energy flux (or intensity) is defined as energy flow per unit cross-sectional area per unit time: $I = c\bar{u} = c\frac{1}{2}\epsilon_0\omega^2 A_0^2$, where $\bar{u}$ is time-averaged energy density. Show that the absorption cross-section per unit solid angle is

$$\frac{d\sigma_{abs}}{d\Omega} = \frac{\hbar\omega}{I} \frac{d\Gamma}{d\Omega} = 32\alpha a_h^2 (k_f a_0)^3 \frac{\sin^2 \theta \cos^2 \phi}{(k_f^2 a_0^2 + 1)^4},$$

where $a_h^2 = \hbar/(m\omega)$ and α is the fine structure constant.

2. Consider a **free** electron interacting with a single photon of energy $\hbar\omega$. For the time being, assume a free electron can absorb or emit a photon. The initial and final wave functions can be taken to be $\psi_i \sim e^{i\mathbf{k}_i \cdot \mathbf{r}}$ and $\psi_f \sim e^{i\mathbf{k}_f \cdot \mathbf{r}}$, respectively. Here $\mathbf{k}_i$ and $\mathbf{k}_f$ are the initial and final wavevectors of the free electron.

Using the most general expression of the transition rates for the absorption and emission, show that a free electron can not absorb or emit a photon. [Hint: this problem has been discussed in Zettili's book]

3. **Equivalence between “potential picture” and “field picture” of the atom-field interaction.** Here “atom-field” interaction means interaction between an atom and an electromagnetic field. The vector potential responsible for the spontaneous emission is given by

$$\mathbf{A}(\mathbf{r}, t) = \sqrt{\frac{2\hbar}{V\epsilon_0\omega}} \frac{1}{2} \sum_{p=1,2} \boldsymbol{\epsilon}_p e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)}.$$

Within the dipole approximation $e^{-i\mathbf{k} \cdot \mathbf{r}} \sim 1$, the vector potential is reduced to

$$\mathbf{A}(\mathbf{r}, t) = \sqrt{\frac{\hbar}{2V\epsilon_0\omega}} \sum_{p=1,2} \boldsymbol{\epsilon}_p e^{i\omega t}.$$

The corresponding electric field is

$$\mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} = -i\sqrt{\frac{\hbar\omega}{2V\epsilon_0}} \sum_{p=1,2} \boldsymbol{\epsilon}_p e^{i\omega t}$$

and the magnetic field is $\mathbf{B}(\mathbf{r}, t) = 0$.

(a) The atom-field interaction term in terms of vector potential is given by

$$H_{\text{int}}^{\text{pot}} = \frac{e}{m} \mathbf{A}(\mathbf{r}, t) \cdot \mathbf{p},$$

where $\mathbf{A}(\mathbf{r}, t)$ is the vector potential generating both the electric and magnetic fields, whereas the momentum operator $\mathbf{p}$ is acting on the atomic states.

Within the first-order time-dependent perturbation theory, show that the total spontaneous emission rate for unpolarized radiation is

$$\widetilde{W}_{\text{spon}} = \frac{4\alpha}{3c^2} \omega_{if}^3 |\mathbf{r}_{fi}|^2,$$

where $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ is the fine structure constant.

- (b) The atom-field interaction term H_{int} can also be expressed in terms of the electric field, instead of vector potential, as

$$H_{\text{int}}^{\text{field}} = e\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, t),$$

where the electric field is $\mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}}{\partial t}$.

Calculate the emission rate using $H_{\text{int}}^{\text{field}}$ and compare with the previous result.

4. Consider an electron ($q = -e$) is trapped in a one-dimensional harmonic oscillator potential $V(x) = \frac{1}{2}m_e\omega_0^2x^2$. Assuming that the electron is initially in the excited state $|n\rangle$ ($n > 0$).

- (a) Obtain the spontaneous emission rate for the transition from the excited state $|n\rangle$ to the appropriate lower state. Note that the emitted photon has the same frequency as the oscillator.

Answer:

$$W_{\text{spon}} = \frac{4n\alpha\hbar\omega_0^2}{6m_e c^2} = \frac{ne^2\omega_0^2}{6\pi\epsilon_0 m_e c^3}.$$

- (b) Show that the radiated power is

$$P_n = \hbar\omega_0 W_{\text{spon}} = \frac{e^2\omega_0^2(n\hbar\omega_0)}{6\pi\epsilon_0 m_e c^3} = \frac{e^2\omega_0^2}{6\pi\epsilon_0 m_e c^3} [(E_n - \hbar\omega_0/2)].$$

Comment on the radiated power P_n if the electron is in the ground state $|n = 0\rangle$.

It can be seen that a quantum oscillator radiates in an almost exactly analogous manner to the equivalent classical oscillator. The only difference is the factor $\hbar\omega_0/2$, the zero-point energy is needed to ensure that the ground-state of the quantum oscillator does not radiate.

- (c) **Comparing with the classical case:** According the Larmor formula in classical electrodynamics, the instantaneous power radiated by an oscillating electron is

$$P_{\text{cl}}(t) = \frac{e^2 a^2(t)}{6\pi\epsilon_0 c^3},$$

where $a(t)$ is the acceleration. Show that the time-averaged power radiated by the electron with energy E is

$$\tilde{P}_{\text{cl}} = \frac{e^2\omega_0^2}{6\pi\epsilon_0 m_e c^3} E.$$

Compare the quantum result with that of the classical result.

- (d) Estimate the spontaneous emission rate and life time of the excited state if $|n = 3\rangle$ for $\omega_0 = 10^{14}$ rad/sec.

- (e) Check whether the dipole approximation is valid or not in this problem.
5. Consider the transition from $2p$ states to $1s$ state in a hydrogen atom. Assuming all the three sub-levels of $2p$ state are equally populated. Total spontaneous emission rate can be written as

$$W_{\text{spont}} = \frac{4\alpha}{3c^2} \omega_{if}^3 |\mathbf{r}_{fi}|^2 \rightarrow \frac{4\alpha}{3c^2} \omega_{if}^3 \frac{1}{3} \sum_{m=0,\pm 1} |\mathbf{r}_{fi}|^2.$$

Here $\alpha = e^2/(4\pi\epsilon_0\hbar c)$, $|i\rangle = |2, 1, m\rangle$ with $m = 0, \pm 1$ and $|f\rangle = |1, 0, 0\rangle$. Calculate

$$\widetilde{W}_{\text{spont}} = \frac{4\alpha}{9c^2} \omega_{if}^3 \sum_{m=0,\pm 1} |\mathbf{r}_{fi}|^2.$$

Hint:

$$|\hat{\mathbf{r}}_{fi}|^2 = |\langle f | \hat{\mathbf{r}} | i \rangle|^2 = |\langle f | \sin \theta \cos \phi | i \rangle|^2 + |\langle f | \sin \theta \sin \phi | i \rangle|^2 + |\langle f | \cos \theta | i \rangle|^2.$$

6. The vector potential operator $\mathbf{A}(\mathbf{r}, t)$ for a monochromatic electromagnetic field with frequency ω confined in cube of volume $V = L^3$ is given by

$$\mathbf{A}(\mathbf{r}, t) = \sqrt{\frac{2\hbar}{V\epsilon_0\omega}} \frac{1}{2} \sum_{p=1,2} \boldsymbol{\epsilon}_p \left[a e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)} + a^\dagger e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)} \right].$$

Calculate the electric field operator $\mathbf{E}(\mathbf{r}, t)$ and the magnetic field operator $\mathbf{B}(\mathbf{r}, t)$.